PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Table of Contents

1. Project Overview3**Error! Bookmark not defined.**2. Background & Business Objective443. Product Purpose444. System Functionalities555. User Stories & Use Cases886. Features Included (Scope Definition)11117. Features Not Included (Scope Control)11118. Assumptions & Constraints12**Error! Bookmark not defined.**9. Risks & Questions Throughout Development 113

Project Name: BudgetBrain: GLM-Powered Financial Decision Assistant

Version: 1.0

1. Project Overview

- **Problem Statement:** Individuals and young professionals in Malaysia struggle to make informed financial decisions due to fragmented data across multiple income sources (allowance, passive income, investments) and unclear future goals. Most people track spending manually but cannot connect their daily spending to long-term goals or optimize their income sources effectively.
- **Target Domain:** Domain 2: AI for Economic Empowerment & Decision Intelligence
- **Proposed Solution Summary:** BudgetBrain removes the friction of financial decision-making by enabling users to input their monthly spending template, monthly allowance, passive income, investments, and future goals. Using Z.AI's GLM, the system analyzes all data and provides: Recommendation, Reasoning, Expected Impact, Confidence, and a Reminder. This shifts users from "guessing about money" to "making informed decisions connected to their goals."

2. Background & Business Objective

- **Background of the Problem:** For many years, personal finance management has followed a fragmented pathway. Users track spending in apps or spreadsheets, but these tools do NOT connect spending to income sources (allowance, passive income) or long-term goals (buying a house, saving for vacation). Users have multiple income streams but no unified decision intelligence. According to studies, people with clear financial goals save 2x more than those without - but most lack the bridge between daily spending and future goals.

- **Importance of Solving This Issue:** Accelerating the time from "knowing spending" to "making goal-aligned decisions" from weeks to seconds. By providing AI-powered decision intelligence that includes REMINDERS for long-term accountability, users can optimize their finances without needing financial expertise. This directly supports economic empowerment.

- **Strategic Fit / Impact:** Aligns with UMHackathon 2026 Domain 2's goal of using AI for economic empowerment. The system demonstrates decision intelligence (not just automation) with quantifiable impact and includes a unique Reminder feature for sustained behavior change.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-powered financial decision partner that analyzes spending, income sources (allowance + passive income), investments, and future goals - then delivers actionable recommendations with reminders for accountability.

3.2. Intended Users (Target Audience):

User Type	Description
Young Professionals	Have salary + side income, want to optimize spending and investments
University Students	Have monthly allowance, want to save for goals (e.g., new laptop, travel)
Anyone with passive income	Want to save money but lack financial analysis skills
Gig Workers	Want to maximize returns and reinvest wisely

Example Persona:

Aisha, 22, university student. Gets RM 800 monthly allowance. Has RM 200 passive income from online tutoring. Wants to save RM 2000 for a new laptop in 5 months. Currently spends without tracking. Needs simple, actionable advice with reminders to stay on track.

4. System Functionalities

4.1. Description: The system operates as a decision intelligence engine powered by Z.AI's GLM. Users input 5 categories:

1. Template monthly spending (e.g., Food, Transport, Entertainment, etc.)
2. Monthly allowance
3. Passive income
4. Investment details
5. Future goal (e.g., "Save RM 2000 for laptop in 5 months")

GLM analyzes all inputs and returns 5 outputs:

1. Recommendation - What action to take
2. Reasoning - Why this recommendation makes sense
3. Expected impact - Quantifiable result (RM saved, time to goal)
4. Confidence - High/Medium/Low
5. Reminder - A specific accountability prompt

4.2. Key Functionalities:

- **Monthly Spending Template: User enters their typical monthly spending categories and amounts**
- **Monthly Allowance Input:** User enters their regular monthly allowance/main income
- **Passive Income Input:** User enters any additional income (rental, dividends, side gigs)
- **Investment Input:** User enters investment details (stocks, crypto, savings, etc.)
- **Future Goal Input:** User enters goal (e.g., "Save RM 5000 for down payment in 8 months")
- **GLM Decision Generation:** System calls Z.AI GLM API with constructed prompt containing all 5 inputs
- **5-Output Display:** Shows Recommendation, Reasoning, Expected Impact, Confidence, Reminder
- **Decision Tracking:** Accept/Reject buttons to record user follow-through

4.3. AI Model & Prompt Design

4.3.1. **Model Selection:** The system uses Z.AI GLM (General Language Model) as required by UMHackathon 2026 rules. GLM is chosen specifically for its:

- Multi-factor reasoning (analyzes spending + multiple income sources + goals together)
- Structured output generation (5 consistent sections)
- Natural language explanation (users need to understand WHY)
- Ability to generate personalized reminders

4.3.2. **Prompting Strategy:** The team uses **few-shot prompting** with explicit 5-section format requirements.

4.3.3. **Context & Input Handling**:

- **Maximum input size:** 1500 tokens
- **What happens when exceeded:** Truncate by summarizing spending categories
- **Input types:** Structured numbers + text goal description

4.3.4. **Fallback & Failure Behavior:**

- **Hallucinated response:** Validate all 5 sections exist. If missing, retry once with temperature=0.1
- **API timeout (10 seconds):** Show "GLM is analyzing. Please try again."
- **Invalid format after retry:** Show "Unable to generate - please check your inputs"

5. User Stories & Use Cases

- **User Stories:**

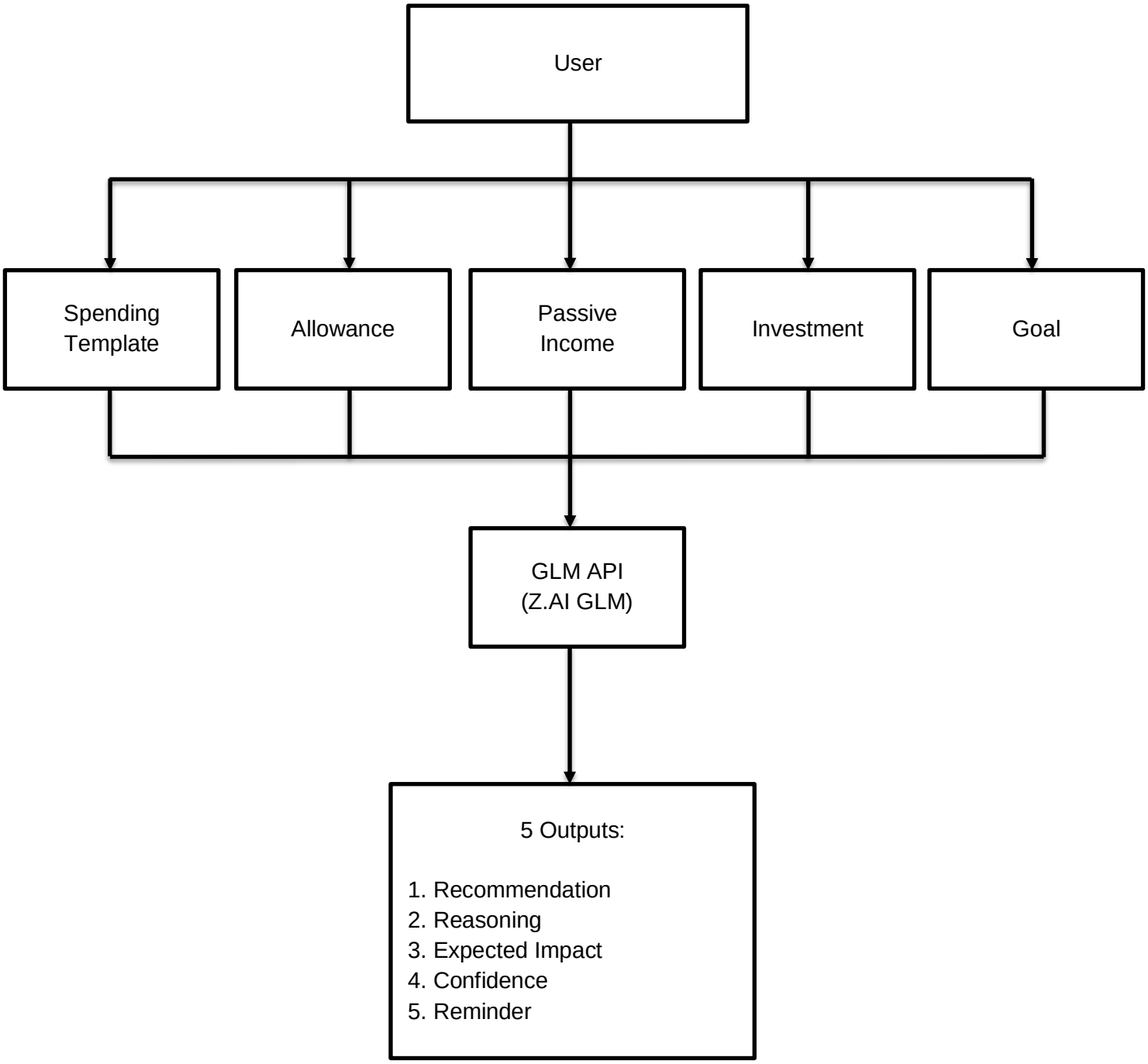
ID	User Story
US-01	As a student, I want to enter my allowance and spending so GLM tells me how to save for my goal
US-02	As someone with passive income, I want GLM to recommend how to reinvest it
US-03	As a goal-oriented user, I want a reminder to keep me accountable
US-04	As a judge, I want to see all 5 outputs (including Reminder) to verify GLM's capability
US-05	As a user, I want to see expected impact in RM so I know if the advice is worth following

- **Use Cases (Main Interactions):**

Input to Decision Flow: User enters monthly spending template (Food: RM 400, Transport: RM 150, Entertainment: RM 200, Shopping: RM 150, Bills: RM 100), monthly allowance (RM 800), passive income (RM 200 from online tutoring), investments (RM 500 in savings account), and future goal ("Save RM 2000 for new laptop in 5 months"). The system constructs a prompt with all 5 inputs, calls GLM API, and returns 5 outputs: Recommendation, Reasoning, Expected Impact, Confidence, and Reminder.

Goal-Based Recommendation Generation: User updates their future goal (e.g., from "laptop" to "down payment for house"). The system re-analyzes with the same spending and income data but the new goal. GLM generates a new recommendation tailored to the changed goal. The Reminder output updates to reflect the new goal timeline (e.g., "Set a monthly transfer of RM 300 to your savings account").

- **Use Case Diagram:**



6. Features Included (Scope Definition)

Feature	Description
Monthly Spending Template	User enters spending categories and amounts
Monthly Allowance Input	User enters main monthly income
Passive Income Input	User enters additional income sources
Investment Input	User enters investment details
Future Goal Input	User enters goal description and timeline
GLM API Integration	Calls Z.AI GLM with all 5 inputs
5-Output Display	Shows Recommendation, Reasoning, Impact, Confidence, Reminder
Accept/Reject Tracking	Records user decision
Without GLM Comparison	Proves GLM is critical

7. Features Not Included (Scope Control)

- Database storage (no user accounts)
- Bank API integration
- Automatic expense tracking
- Investment performance calculations (GLM provides recommendations only)
- Multi-user collaboration
- SMS/Email notifications (reminder is displayed, not sent)
- Historical trend analysis

8. Assumptions & Constraints

- **LLM Cost Constraint:** The estimated token cost per user session is approximately 1000 to 1400 tokens, which includes both the prompt and the 5-section response from GLM. To manage costs effectively, the team has implemented a design decision to use a fixed-length prompt and maintain no conversation history between sessions. This ensures that each API call is independent and token usage remains predictable.
- **Technical Constraints:** The system is built exclusively using the Streamlit web framework for rapid prototyping. For the preliminary round, the application will be deployed locally on a development machine rather than on cloud infrastructure. Additionally, the MVP does not include a database; all user data and session state are stored temporarily in-memory using Streamlit's session state feature.
- **Performance Constraints:** The target response time for GLM API calls is less than 8 seconds under normal conditions. A timeout limit of 10 seconds has been set to prevent indefinite waiting. If the GLM API does not respond within 10 seconds, the system will display a friendly error message and allow the user to retry.
- **User Input:** All spending amounts entered by the user must be greater than or equal to zero. Similarly, the monthly allowance field also requires a value of zero or higher. The future goal field must contain a text description and cannot be left empty.

9. Risks & Questions Throughout Development

- Risk 1: GLM Forgets the Reminder Section

  Question: How do we ensure GLM consistently outputs all 5 sections?

  Mitigation: Response validation checks for "Reminder:" and retries once with lower temperature if missing.

- Risk 2: GLM Provides Generic Recommendations

  Question: How do we ensure recommendations are specific to the user's data?

  Mitigation: Prompt includes exact spending amounts, income sources, and goal. Few-shot examples show personalization.

- Risk 3: User's Goal is Too Vague

  Question: What happens if the goal is unclear (e.g., "I want to be rich")?

  Mitigation: GLM is instructed to ask clarifying questions for specific amount and timeline.

- Risk 4: GLM Hallucinates Numbers

  Question: How do we prevent incorrect financial calculations?

  Mitigation: GLM provides directional recommendations; exact impacts are calculated by the system where possible.

- Risk 5: No GLM API from Organizers

  Question: What if API credentials are never provided?

  Mitigation: Complete mock implementation ready, clearly labeled as demonstration mode.

- Risk 6: Response Time Too Slow

  Question: What if GLM takes more than 10 seconds?

  Mitigation: 10-second timeout with loading spinner and retry option.